

Blood Bank Web Application Development

Hi

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Introduction: Overview of the Blood BankWEB Application

➡ Our web application aims to streamline blood donation and transfusion processes, enhancing efficiency and accessibility for both donors and recipients.

➡ The system will provide a user-friendly interface for donors to schedule appointments, track their donation history, and access educational resources.



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Problem Statement

Inefficient Processes

Current manual systems often lead to delays, inaccuracies, and difficulties in managing blood inventory.

Limited Access to Information

Lack of centralized data and real-time updates hinder effective decision-making and resource allocation.

Security Concerns

Sensitive medical data requires robust security measures to protect patient privacy and ensure³ data integrity.



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An example of blood bank link is given below

https:

//eraktkosh.mohfw.gov.in/BLDAHIMS/bloodbank/transactions/bbpublicindex.html

Checking this site we can make few improvements



Modernize Design:

- ➡ Replace the outdated layout with a sleek, responsive design for improved usability on mobile devices.
- ➡ Use clean fonts, better spacing, and high-quality visuals to create a polished look.

•Locate Nearby Blood Banks:

- ➡ Use Google Maps API or OpenStreetMap to display nearby blood banks.
- ➡ Allow users to input a location and filter blood banks by distance or availability.

•Real-time Notifications:

- ➡ Notify users of emergency requirements or donation camps using geolocation push notifications .
- ➡ Integrate geofencing to alert users when near a blood bank.



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Key Features and Functionality

Donor Management

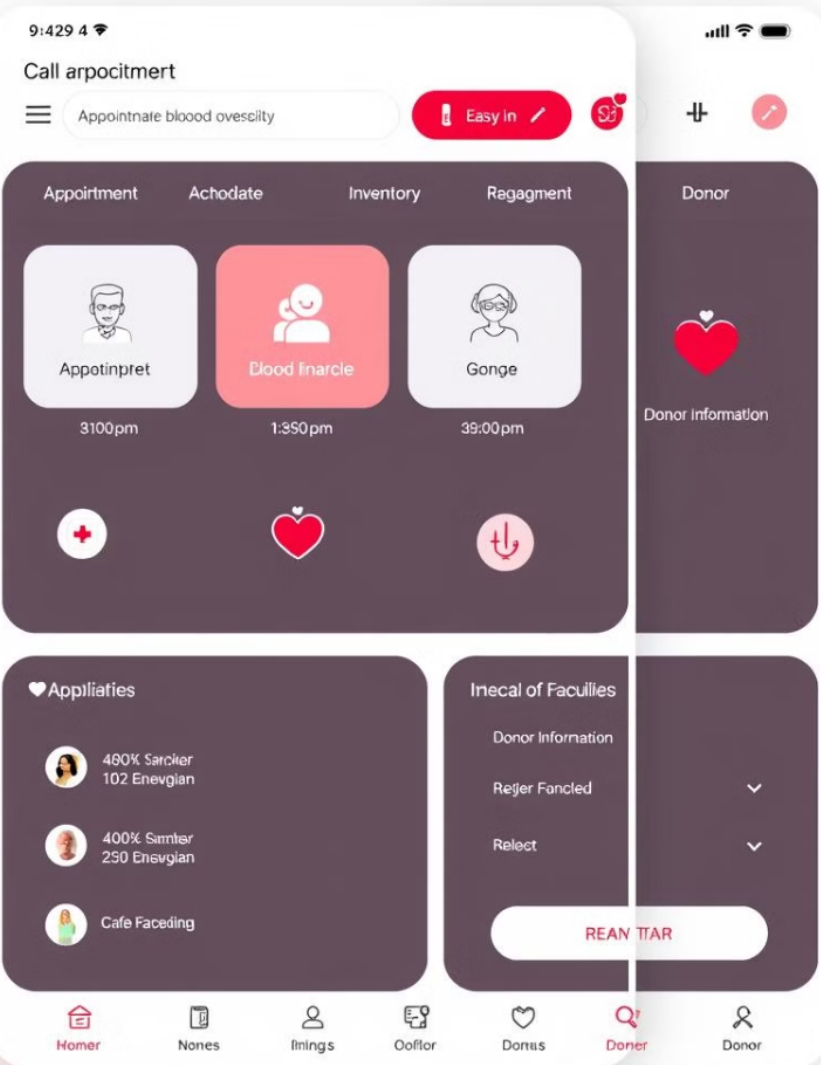
This feature allows users to register, schedule appointments, track their donation history, and access educational materials.

Blood Inventory Management

This module enables hospitals to manage blood inventory, track blood types, expiration dates, and order blood units.

Transfusion Management

This feature facilitates secure blood transfusion processes, including blood typing, cross-matching, and documentation.



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TOOLS USED

Front-end

React, HTML, CSS, JavaScript

Back-end

Node.js, Express.js, MongoDB

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def __init__(self, settings):
    self.file = None
    self.fingerprints = set()
    self.logdupes = True
    self.debug = debug
    self.logger = logging.getLogger(__name__)
    if path:
        self.file = open(os.path.join(path, 'requests.log'),
                        'a')
        self.file.seek(0)
        self.fingerprints.update(self._get_fingerprints())

    @classmethod
    def from_settings(cls, settings):
        debug = settings.getbool('DEBUG', False)
        return cls(job_dir(settings), debug)

    def request_seen(self, request):
        fp = self.request_fingerprint(request)
        if fp in self.fingerprints:
            return True
        self.fingerprints.add(fp)
        if self.file:
            self.file.write(fp + os.linesep)

    def request_fingerprint(self, request):
        return request_fingerprint(request)
```



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Backend Development and API Integration



Cloud Hosting

Deploy the application on a reliable cloud platform, ensuring scalability and accessibility.

Eg: AWS



Database Integration

Integrate the application with the chosen database, implementing secure data access and retrieval mechanisms.

Eg : MySql



API Integration



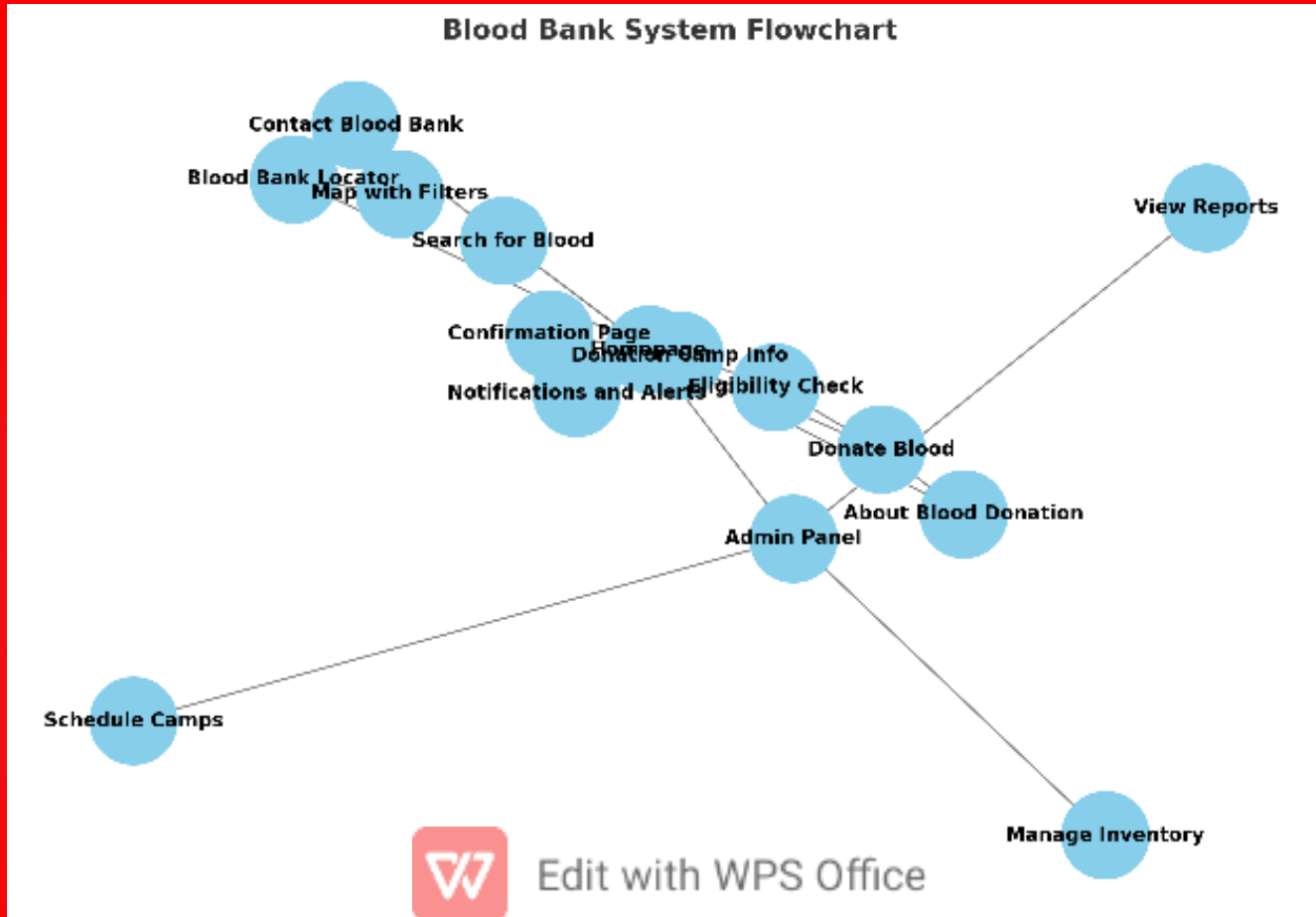
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DFD level1



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FLOWCHART



THANK YOU



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